DSA SCHEDULE

Beginner-Calibrated Curriculum Roadmap (OOPs to Segment Trees) | CAT-1, CAT-2 & TEE Aligned

Nov 05, 2026 – Jun 20, 2027

228 Days	18 Modules	148 Days	43 Days	37 Days
TOTAL SPAN	FULL CURRICULUM	CORE PREP	DEDICATED REVISION	EXAM WINDOWS (3X)

MODULE & TOPIC FOCUS	DIFFICULTY	CORE PREP	REVISION	TOTAL	CALENDAR WINDOW	CORE BEGINNER STRATEGY
21. OOPs in Java	Medium	5 Days	2 Days	7 Days	Nov 05 – Nov 11, 2026	Classes, objects, inheritance, runtime polymorphism, and encapsulation.
22-23. Recursion (Part 1 & 2)	Hard	10 Days	3 Days	13 Days	Nov 12 – Nov 24, 2026	Stack frames activation, recursion tree drawing, recurrence relations.
24. Divide & Conquer	Medium	4 Days	1 Days	5 Days	Nov 25 – Nov 29, 2026	Merge and Quick Sort index division array handling & partition math.
25. Time & Space Complexity	Easy-Med	3 Days	1 Days	4 Days	Nov 30 – Dec 03, 2026	Big-O asymptotic bounds, recursive call stack space, loop equations.
26. Backtracking	Hard	6 Days	2 Days	8 Days	Dec 04 – Dec 11, 2026	State-space trees, branch pruning, N-Queens, Sudoku solver logic.
27. ArrayLists	Easy	4 Days	1 Days	5 Days	Dec 12 – Dec 16, 2026	Dynamic sizing mechanics, multi-dimensional lists, container problems.
🛑 CAT-1 Examination Window (LOCKED)				8 Days	Dec 17 – Dec 24, 2026	0 DSA Days — 100% University Exam Focus & GPA Shield.
28-29. Linked Lists (Part 1 & 2)	Med-Hard	10 Days	3 Days	13 Days	Dec 25 – Jan 06, 2027	Pointer changes, reversals, Floyd's cycle detection, Merge Sort on LL.
30-31. Stacks (Part 1 & 2)	Med-Hard	9 Days	3 Days	12 Days	Jan 07 – Jan 18, 2027	Monotonic stacks, Next Greater Element, Largest Rectangle in Histogram.
32. Queues	Medium	5 Days	1 Days	6 Days	Jan 19 – Jan 24, 2027	FIFO dynamic buffer loops, Deques, and queue generation via stacks.
33. Greedy Algorithms	Medium	5 Days	2 Days	7 Days	Jan 25 – Jan 31, 2027	Fractional Knapsack, activity selection, interval heuristics.
34-36. Binary Trees (Part 1, 2 & 3) Pre-CAT-2 Block	Hard	16 Days	4 Days	4 Days	Feb 01 – Feb 04, 2027	Traversals (DFS/BFS), diameter, subtree matching, vertical views.
🛑 CAT-2 Examination Window (LOCKED)				8 Days	Feb 05 – Feb 12, 2027	0 DSA Days — 100% University Exam Focus & GPA Shield.
34-36. Binary Trees (Part 1, 2 & 3) Post-CAT-2 Block	Hard	16 Days	4 Days	16 Days	Feb 13 – Feb 28, 2027	Traversals (DFS/BFS), diameter, subtree matching, vertical views.
37-38. Binary Search Trees (Part 1 & 2)	Medium	7 Days	2 Days	9 Days	Mar 01 – Mar 09, 2027	BST validation traits, structure-specific deletions, balanced BSTs.
39. Heaps / Priority Queues Pre-TEE Block	Med-Hard	6 Days	2 Days	1 Days	Mar 10 – Mar 10, 2027	Binary heap design, heapify algorithm, k-way merges.
🛑 TEE Examination Window (LOCKED)				21 Days	Mar 11 – Mar 31, 2027	0 DSA Days — 100% University Exam Focus & GPA Shield.
39. Heaps / Priority Queues Post-TEE Block	Med-Hard	6 Days	2 Days	7 Days	Apr 01 – Apr 07, 2027	Binary heap design, heapify algorithm, k-way merges.
40. Hashing	Medium	8 Days	2 Days	10 Days	Apr 08 – Apr 17, 2027	Collision chaining, internal HashMap architecture, sliding windows.
41. Tries	Medium	7 Days	2 Days	9 Days	Apr 18 – Apr 26, 2027	Prefix tree construction, startsWith queries, autocomplete search models.
42-47. Graphs (Part 1-5 + Supplemental)	Very Hard	21 Days	5 Days	26 Days	Apr 27 – May 22, 2027	Adjacency structures, BFS/ DFS, Dijkstra, Bellman-Ford, Prim's, DSU.
48-53. Dynamic Programming (Part 1-6)	Very Hard	18 Days	4 Days	22 Days	May 23 – Jun 13, 2027	Memoization to Tabulation, 0-1 Knapsack, LCS, LIS, Catalan Numbers.

MODULE & TOPIC FOCUS	DIFFICULTY	CORE PREP	REVISION	TOTAL	CALENDAR WINDOW	CORE BEGINNER STRATEGY
54. Segment Trees	Hard	5 Days	2 Days	**7 Days**	`Jun 14 – Jun 20, 2027`	Range sum/min queries, point updates, lazy propagation mechanisms.

Academic Policy Note: CAT-1 (8 Days), CAT-2 (8 Days), and TEE (21 Days) have zero DSA tasks allocated to protect your GPA. Curriculum pauses and resumes without context loss.